

From $J_3(\mathbb{O})$ to $E_7(-25)$: The Kantor-Koecher-Tits Construction, Computationally Verified

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Abstract. A companion paper constructs $J_3(\mathbb{O})$ and its 52-dimensional derivation algebra $\mathfrak{f}_4$ from scratch, verifying both directly rather than assuming them from citation. Here we extend that construction to the full 133-dimensional simple Lie algebra given by the Kantor-Koecher-Tits (TKK) construction, using $\mathfrak{str}(J_3(\mathbb{O})) = \mathfrak{der}(J_3(\mathbb{O})) \oplus L(J_3(\mathbb{O}))$ as the grade-zero part. A first, plausible bracket formula fails the Jacobi identity outright; a precisely sourced correction, verified independently rather than trusted, satisfies it exactly, to machine precision, across every case tested. Diagonalizing the resulting 133×133 Killing form directly gives signature exactly -25 : the algebra is $\mathfrak{e}_{7(-25)}$, the real form associated with "magic" $\mathcal{N} = 2$ Maxwell-Einstein supergravity (Günaydin-Sierra-Townsend, 1983) — and not the maximally split $\mathfrak{e}_{7(7)}$ familiar as the U-duality algebra of $\mathcal{N} = 8$ supergravity, a distinction fixed by the computation rather than by dimension counting, since both share dimension 133. The 28-dimensional $\mathfrak{so}(8)$ frame stabilizer from the companion paper is confirmed, directly against the full 133-dimensional bracket, to sit cleanly within the grade-zero subalgebra. This note is confined to the verified construction; physical interpretation beyond the real-form identification is deliberately left for separate, future treatment.

1. Introduction

A companion paper ($*J_3(\mathbb{O})$: Construction, the Peirce Split, and Verified Frame Relativity via F_4^*) builds the exceptional Jordan algebra and its 52-dimensional derivation algebra $\mathfrak{f}_4$ from raw octonion multiplication, verifying every defining property directly. The Kantor-Koecher-Tits construction extends this further: from any Jordan algebra J , it builds a 3-graded simple Lie algebra $\mathfrak{g} = J \oplus \mathfrak{str}(J) \oplus J$, where $\mathfrak{str}(J) = \mathfrak{der}(J) \oplus L(J)$. For $J_3(\mathbb{O})$, this gives $27 + 79 + 27 = 133$ — matching the dimension of the exceptional Lie algebra $\mathfrak{e}_7$. Which of $\mathfrak{e}_7$'s several inequivalent real forms results is not fixed by dimension alone: $\mathfrak{e}_{7(7)}$ (maximal compact subalgebra $\mathfrak{su}(8)$) and $\mathfrak{e}_{7(-25)}$ (maximal compact subalgebra $\mathfrak{e}_6 \oplus \mathfrak{u}(1)$) share dimension 133 but are genuinely different Lie algebras, and the two correspond to different physical theories in the supergravity literature. Determining which form the construction actually yields therefore requires computing an invariant that distinguishes them — the Killing form signature — rather than counting dimensions. This note constructs the TKK algebra from $J_3(\mathbb{O})$ explicitly and determines, by direct computation, which real form results.

2. $\mathfrak{str}(J_3(\mathbb{O}))$: verified, 79-dimensional

$L(J) = \{L_a : a \in J\}$, the space of Jordan-multiplication operators, was combined with the 52 verified $\mathfrak{f}_4$ generators to give 79 candidate generators for $\mathfrak{str}(J)$. These were confirmed linearly independent (rank 79) and closed under the commutator bracket, checked exhaustively across all 3081 pairs with zero violations.

3. A failed first attempt, and why it matters

An initial TKK bracket was implemented from a plausible, standard-looking formula: for (a_1, T_1, b_1) and (a_2, T_2, b_2) ,

$$(T_1(a_2) - T_2(a_1), [T_1, T_2] + \theta(a_1, b_2) - \theta(a_2, b_1), T_1^*(b_2) - T_2^*(b_1)),$$

with $\theta(a, b) = L_{a \circ b} + [L_a, L_b]$ and T^* a sign-involution on $\mathfrak{str}(J)$. Tested against the Jacobi identity — the defining property of any Lie algebra bracket — this failed outright: residuals of 94 to 130 across independent random trials, not a marginal numerical discrepancy.

This is reported for the same reason earlier failures in this series were reported rather than silently corrected: a plausible-looking formula, assembled from genuinely correct individual pieces, can still be wrong in its precise combination, and only the defining identity itself reliably catches this.

4. The corrected bracket, precisely sourced

Rather than guess at further corrections, the exact TKK commutation relations were located directly (Meng, *Euclidean Jordan Algebras, Hidden Actions, and J -Kepler Problems*, arXiv:0911.2977, §3, citing Koecher 1967): for the Jordan triple product $\{uvw\} = u(vw) + w(vu) - (uw)v$ and $S_{uv}(z) = \{uvz\}$,

$$[X_u, Y_v] = -2S_{uv}, \quad [S_{uv}, X_z] = X_{\{uvz\}}, \quad [S_{uv}, Y_z] = -Y_{\{vuz\}}.$$

Working through these relations by hand against the initial formula identified two specific errors: a missing factor of -2 in the grade-zero component, and an incorrect sign together with a swapped pairing in the grade (-1) component. The corrected bracket,

$$(T_1(a_2) - T_2(a_1), [T_1, T_2] - 2\theta(a_1, b_2) + 2\theta(a_2, b_1), \text{star}(T_2)(b_1) - \text{star}(T_1)(b_2)),$$

was tested against the Jacobi identity across independent random trials: residuals of exactly 0.00000000, machine precision, in every case.

5. The Killing form: $\mathfrak{e}_{7(-25)}$, computed directly

The full 133×133 Killing form was computed from the verified bracket — not assumed from the grading, and not taken from citation. Diagonalizing directly: 54 positive eigenvalues, 79 negative, signature exactly -25 , dimension summing correctly to 133.

This matches Meng's own table for $\mathcal{H}_3(\mathbb{O})$ (Meng's notation for the same algebra referred to as $J_3(\mathbb{O})$ throughout this paper) ($\mathfrak{co}(V) = \mathfrak{e}_{7(-25)}$) independently of the present computation. **The algebra is $\mathfrak{e}_{7(-25)}$.**

6. The $\mathfrak{so}(8)$ frame stabilizer embeds cleanly

The 28-dimensional frame-stabilizer subalgebra from the companion paper — the subspace of $\mathfrak{f}_4$ under which the full diagonal Jordan frame remains diagonal — was checked directly against the complete 133-dimensional bracket, not merely assumed

closed from the general grading rule $[\mathfrak{g}_0, \mathfrak{g}_0] \subseteq \mathfrak{g}_0$. It closes exactly, and its basis is confirmed to have zero component in both the $+1$ and -1 graded parts: the stabilizer sits entirely within $\mathfrak{g}_0$, as the construction predicts and as was here independently confirmed.

7. Scope

This note is confined to the verified construction and the real-form identification. It deliberately excludes further physical interpretation — including, but not limited to, any claim that W literally is Planck's constant (as opposed to playing a structural, algebraic role analogous to it, already established in the companion paper on algebraic R-flux), any claim that the Inönü–Wigner contraction producing R-flux "flattens" this E_7 structure, and any framing of R-flux as a privileged boundary or "horizon" adjacent to the non-representable side. None of these has been tested; the last is, in a precise sense already established elsewhere in this series, in tension with directly verified results showing R-flux has no special relationship to $\mathbb{O} \cdot e_8$ by any measure checked. Whether R-flux's own $\{X, P\} \sim W$ structure interacts specially with $J_3(\mathbb{O})$'s diagonal/off-diagonal split — the one place in this investigation where genuine, non-generic structure has been found — remains a separate, open, and not-yet-tested question.

References

- Companion papers: $*J_3(\mathbb{O})$: Construction, the Peirce Split, and Verified Frame Relativity via F_4^* ; *R-Flux Does Not Bridge to the Wild Side*; *The Central Element of Algebraic R-Flux*.
- Meng, G., "Euclidean Jordan Algebras, Hidden Actions, and J -Kepler Problems," arXiv:0911.2977.
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